OXFORD
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INTERNATIONAL A-LEVEL PHYSICS

(9630) PH03

Report on the examination

January 2024

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REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH03 JANUARY 2024

This paper was generally accessible to the students and discriminated well, with the mean mark around 38. The cohort performed, on average, slightly worse in the multiple-choice section (section B) than in section A. As in previous series, the cohort answered calculation-style questions generally better than those requiring an explanation. Also, as in previous series, many students seemed ill-prepared when it came to defining quantities or showing full working in their answers.

QUESTION 01

Question 01.1 was a "show that" and students must show all working and give their final answer to at least one more significant figure than is quoted in the question. Several students gave their final answer as $320\ \text{N}$ which meant they could not access the one mark available for this step.

01.2 was a four-mark question in which many students failed to gain marks due to not including working for intermediate steps or, in some cases, not including coherent equations or subsidiary answers that could gain partial credit.

In question 01.3 the majority of students who only gained partial credit on this question did not show full working or failed to write a conclusion after their calculations. Students must show the steps they have taken to reach an answer rather than assuming that the mathematical steps are obvious and therefore do not need to be included.

In 01.4 many students were able to access at least one mark for stating which angle and which tension were larger. Poor explanations let students down in this question.

QUESTION 02

In 02.1 several students labelled their downwards arrow as "G" or "*gravity*" rather than weight. Some students did not draw their vector arrows on the diagram despite being instructed to do so in the question.

In question 02.2 the most common mistake was that students did not recognise that there would be a resultant force upwards that takes into account the weight of the jumper. This allowed access to one of the available marks.

The unit for spring constant in this question was wrongly stated on the question paper as N kg^{-1}. A thorough analysis of the students' responses was carried out which showed that if a student used this unit to determine an equation to use and they carried this through correctly, they could still access both marks. They were not disadvantaged on the following questions due to error carried forward (ecf) marks in their final answers.

The unit has been corrected (N m^{-1}) on the published version of the question paper.

The majority of students were able to access both marks for 02.3 due to error carried forward from 02.2. The only frequent common error was a rounding error, where students rounded a value that was 15.5 to three significant figures to 16 for their final answer but the correct answer rounded to 15 to two significant figures.

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02.4 was, on the whole, very straightforward with most students accessing at least one of the marks and the majority gaining full marks.

In 02.5 many students scored one of the two marks for appreciating that acceleration or force was not proportional to displacement in this case but are required to be for simple harmonic motion. Few students gained the second mark through either route.

In 02.6 some students did not follow the "explain" command word and just stated that the rope would undergo damping. As air resistance is negligible, more detail was required about where and how the energy is lost in the system. Very few students explained that it was elastic potential energy that is converted into heat and instead wrote about energy in general terms which was not enough to gain the mark.

QUESTION 03

In 03.1, as horizontal was stated in the question and the diagram was drawn at an angle, students were expected to define the direction as *vertically* downwards.

03.2 was done well in general with many students covering all three areas in some detail.

In question 03.3 many students did not link the constant frequency of the voltage to their answer and instead tried to connect it to the radius of the proton. Again the "state and explain" command word was ignored by some students who made bald statements which could not access the mark.

03.5 was generally done well; however, some students wrongly used $\text{velocity} = \frac{\text{distance}}{\text{time}}$, assuming that the radius is constant throughout. This did bring them to the correct answer through incorrect physics.

In 03.6 very few students appreciated that the proton would be accelerated through the potential difference twice in each rotation. The most common partial answer was twice that of the final value.

QUESTION 04

Questions 04.1, 04.2 and 04.3 were straightforward calculations and were on the whole done well.

In 04.4 many students did not answer the question and instead gave facts about radioactive decay in general. Very few students discussed collisions and even fewer students linked these collisions with a transfer of kinetic energy. Very few students mentioned the neutrino. Overall, this question was answered very poorly.

QUESTION 05

05.1 was not answered well as many students thought this was a graph showing potential or wrote about positive and negative charge. A large proportion of students described the shape of the graph rather than explaining why it had that shape.

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05.3 seemed to confuse many students. Most understood that all values should be negative. However students, following a poor performance in 05.1, did not seem to understand the significance of what they were plotting.

QUESTION 06

In question 06.1 many did not mention the linkage of the flux and therefore could not access the mark.

Students struggled with 06.2 as they listed features of the system that would cause an increase in flux rather than an explanation of how that links back to the rate of change of flux linkage and, therefore, electromotive force. Some students discussed the general case of what would create a large rate of change of flux linkage, rather than discussing what happened in this system.

06.3 was done rather poorly as students seemed quite unsure what to do. Many students still had the quantity N_S in their final answer meaning that they had missed steps from their solution and therefore could not access the marks.

06.5 was at least partially accessible to some students. The main errors occurred when students did not link the larger current to the greater flux or flux density.

SECTION B

Most of the students answered more than half of the multiple-choice questions correctly. The easiest questions were 9 and 11 where over 80% of the cohort obtained the correct answer.

In all the questions the most popular answer was the correct answer. The questions with significant distractors were 16 and 20. In these questions only around one-third of students selected the right option.

Question 20 proved to be the most challenging question on section B with only 32% of students gaining a mark.

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